



Shane Rowand

Perkasie, PA | (215) 960-5791 | SRow36912@gmail.com | linkedin.com/in/shane-rowand

Mechanical & Robotics Engineering student seeking a Mechanical/Robotics Engineering internship

Education

Rensselaer Polytechnic Institute

Troy, NY

B.S. in Mechanical Engineering & Robotics Engineering — Expected May 2030

- Relevant Coursework: Introduction to Robotics, Computer-Aided Design (Siemens NX), Multivariable Calculus, Linear Algebra, Physics I & II, Computer Science I

Technical Skills

Mechanical Design: Siemens NX, SolidWorks, Fusion 360, OnShape

Fabrication & Manufacturing: 3D printing/prototyping, Tinkercad, woodworking, resin casting, basic electrical wiring

Robotics & Controls: MuJoCo/MJX, reinforcement learning (PPO), URDF, Docker

Programming: Python, self-taught software development

Platforms: Windows, Git, VS Code

Projects

Axle — Legged-Robot Platform & Personal AI Assistant

Personal Project

- Extending an AI agent platform into a hexapod legged robot, architecting a two-layer control system that separates high-level LLM reasoning from a low-level locomotion policy
- Training a reinforcement learning locomotion policy (PPO, domain randomization) in MuJoCo/MJX at 50–200 Hz control rates, developing a single-leg prototype before scaling to the full hexapod
- Modeling leg assemblies in CAD and exporting URDF models to integrate mechanical design directly into the simulation pipeline
- Built the underlying agent core (Python, WebSocket architecture, Docker-sandboxed execution) that serves as the robot's high-level reasoning layer

Deck Reconstruction — Structural Rebuild

Hands-On Project

- Rebuilt the structural framing of a ~250 sq ft deck, replacing rotted and warped joists and boards after identifying load-bearing failure points
- Calculated load distribution and beam span requirements before construction, applying statics and physics coursework to a real-world structural design

Seatbelt Usage Tracker — PennDOT Competition

Prototype

- Engineered a clamp-on mechanical device to track seatbelt buckle usage, addressing a road-safety data problem posed by PennDOT
- Modeled the housing in Tinkercad and 3D printed a working prototype, iterating on fit and function to present at the competition

Custom PC Build & Troubleshooting

Personal Project

- Built and repeatedly rebuilt a custom desktop PC, including multiple custom liquid-cooling loops, CPU/GPU upgrades, overclocking, and power-budget calculations
- Independently diagnosed and resolved hardware and software failures, including a blown motherboard fuse and post-rebuild instability after a full case transfer

Activities & Leadership

RPI Robotics Club

Troy, NY

Member

2026 – Present

- **Combat Robot:** Designing and manufacturing a 1lb combat robot in OnShape; 3D printing, assembling, and wiring all components ahead of an upcoming competition against other teams

Work Experience

Weis Markets

Perkasie, PA

Produce Associate

2024 – Present